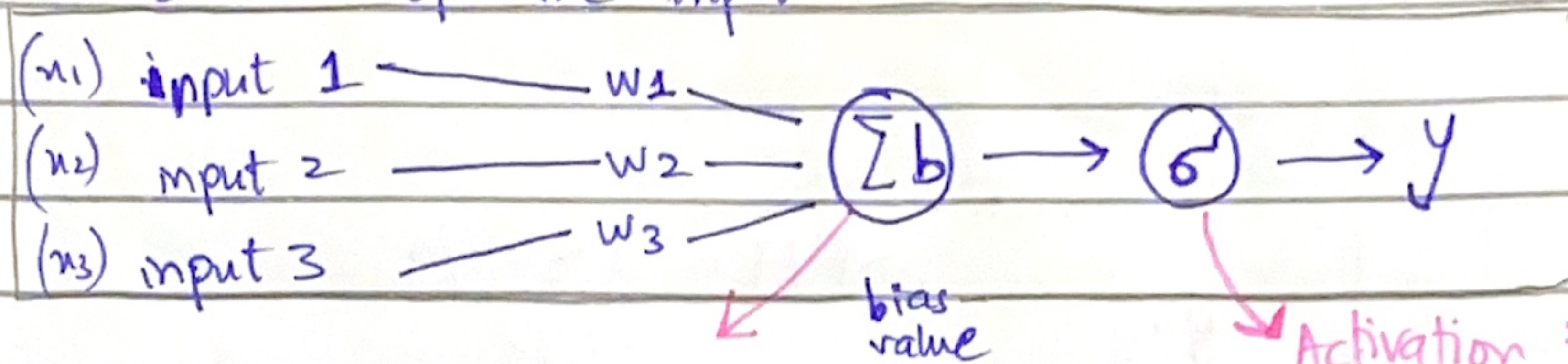


LECTURE-16

Monday Continue
30/3/28

Artificial Neural Network (ANN):
⇒ Classification: Predicts a category.

Perceptron: an ANN model which makes decision on basis of weights and biases upon the input



$$\Sigma b = \Sigma x_i \cdot w_i = x_1 w_1 + x_2 w_2 + x_3 w_3 + b \quad (\text{Linear Combo.})$$

Same as Regression eq: $y = m x + c$. Means w/out activation func; Perceptron is just a regression model.
 Σb ← weights ← inputs ← bias

Activation func: It is a func. that converts a useless value found by Σb into a value that model can plot on graph & see that in which category does the input falls into on basis of threshold set.
 $\therefore y = \sigma(\Sigma b)$ where $\sigma(x)$ is activation func. & 'x' is answer of Σb .

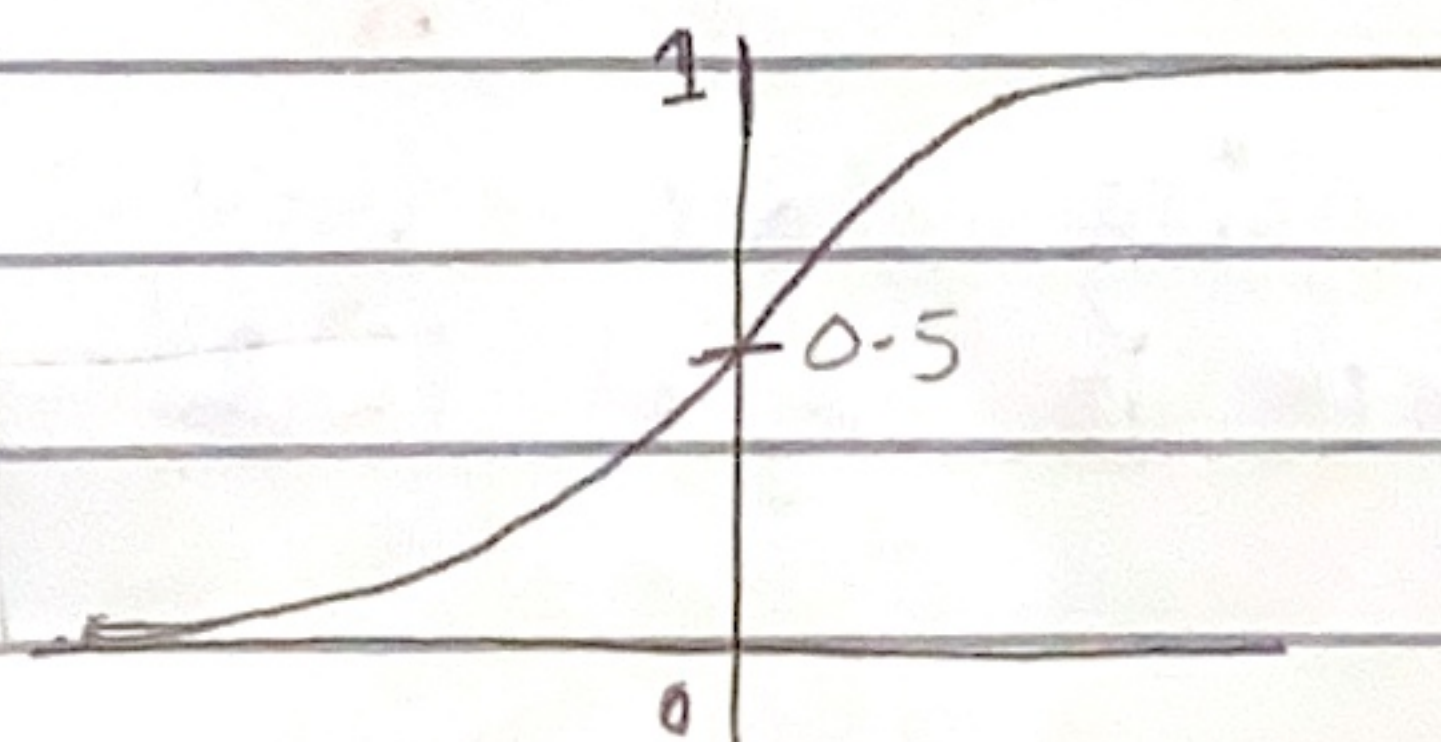
Q: we're input features for a specific group i.e: $\overset{x_1}{0.1}, \overset{x_2}{0.5}, \overset{x_3}{0.3}$.

All $\overset{w_1, w_2, w_3}{\text{weights}} = 0.1$ and bias = 1 with Sigmoid activation func.

Find which group does the features belong to, 0 or 1.

Sigmoid Act. Func: Smashes any input into a range of 0 - 1.

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$



$$\sum b = [(0.1)(0.1) + (0.1)(0.5) + (0.3)(0.1)] + 1$$

$$\sum b = n = \boxed{1.09}$$

* Now this doesn't give any info about group 0 or 1 so put in $\sigma(n)$

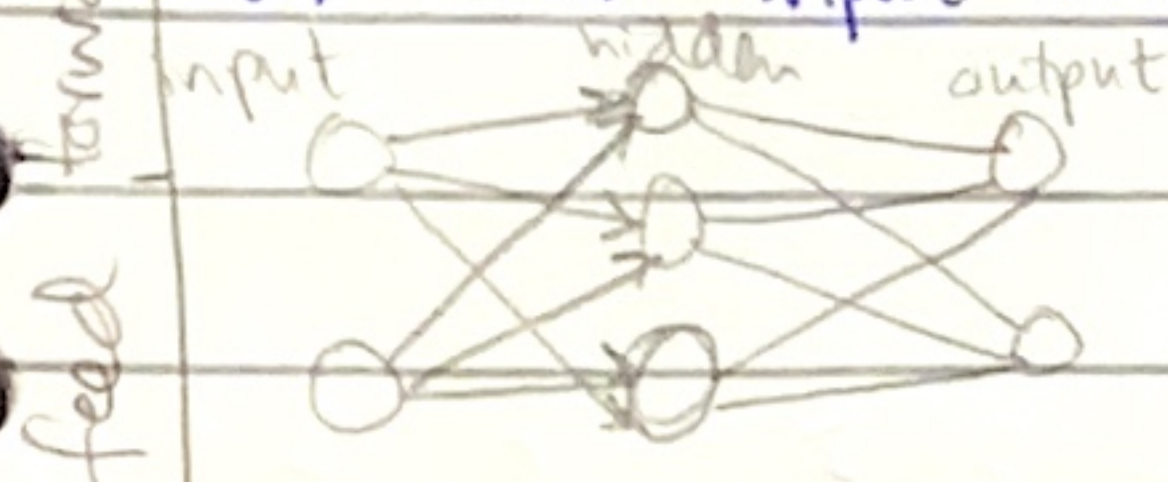
$$\sigma(1.09) = \frac{1}{1 + e^{-1.09}} = \boxed{0.748}$$

answer: since 0.748 is greater than 0.5 (half) $\therefore$ the input whose features are given in ques. belongs to Group 1

$\Rightarrow$ Types of ANN:

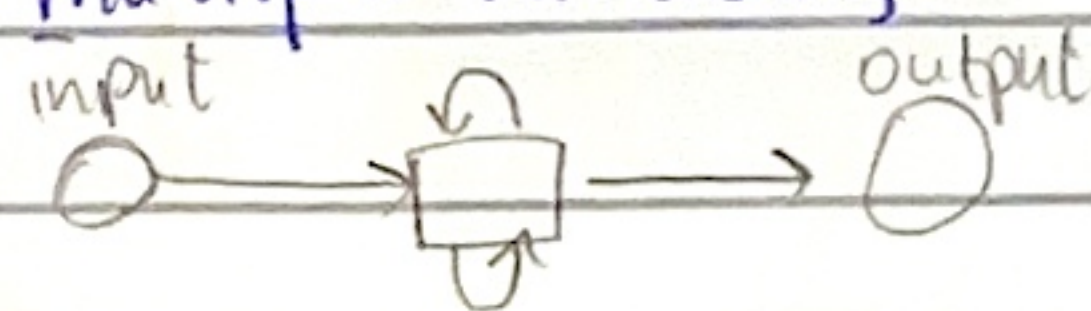
- Single layer Perceptron: one input & one output layer
- Multi layer: 1 input layer, 1 hidden layer and 1 output layer

BUT each input has connection with each neuron ~~in~~ in hidden layer.

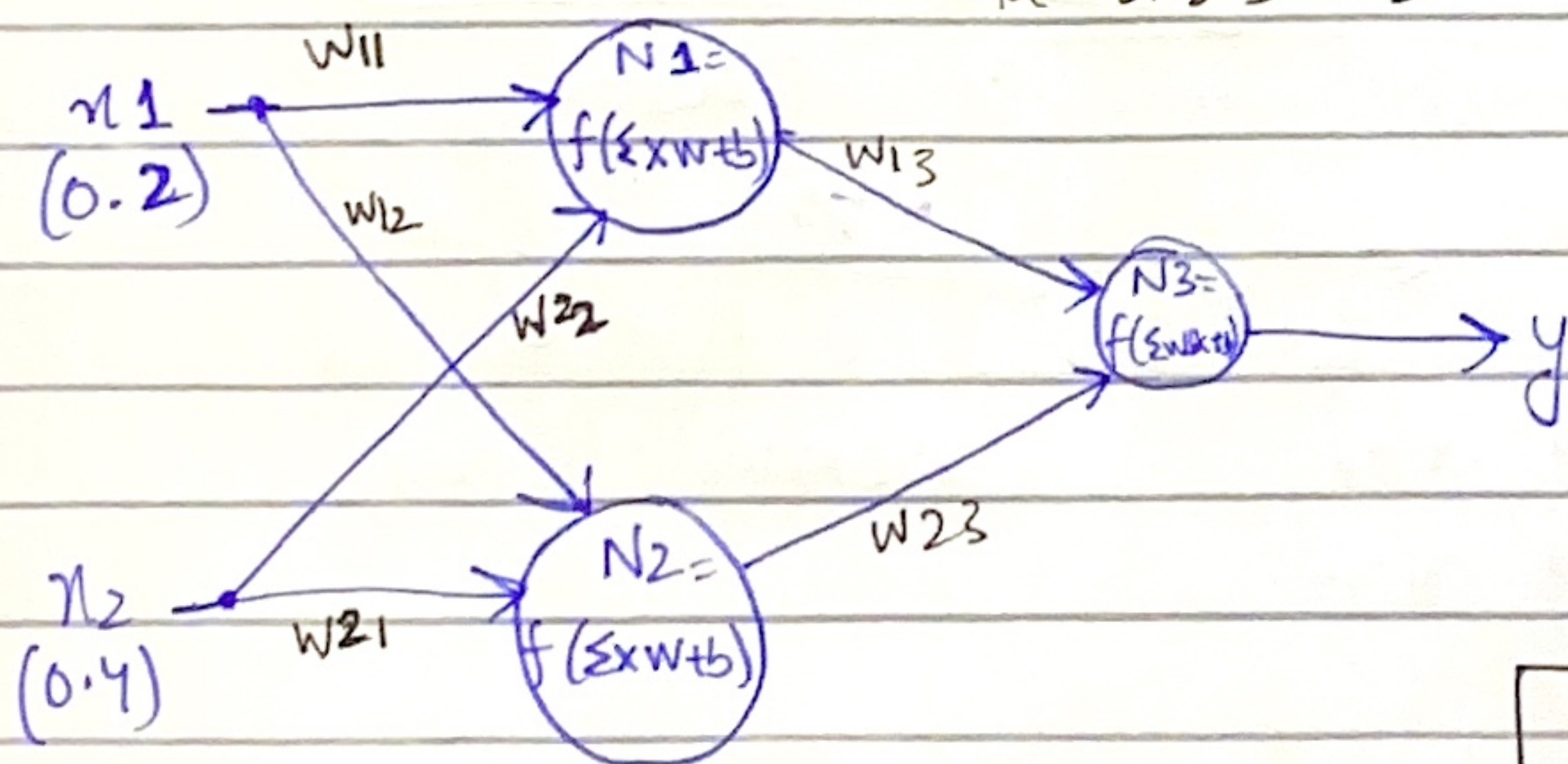


- Deep Learning Network: 1 input, multiple hidden, 1 output.

- Recurrent Neural Network:



$\Rightarrow$ Multi layer ANN: Weights of all $n_2 = 0.2$, Weights of n_1 are all $= 0.1$
All biases $= 1$.



$$N1 = (0.1)(0.2) + (0.1)(0.4) + 1 = 1.06$$

$$N2 = (0.2)(0.2) + (0.2)(0.4) + 1 = 1.12$$

* Pass $N1$ & $N2$ through activation func. sigmoid.

$$\sigma(N1) = 0.742, \sigma(N2) = 0.754$$

$$Y = N3 = [(0.1)(0.742) + (0.2)(0.754)] + 1$$

$$\boxed{Y = 0.89}$$